

BST Vol 3 Ch 5 — Halo Nuclei: Dripline Structure via BST Substrate Extension (v0.3, Wave 1)

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Vol 3 Chapter 5 — Halo Nuclei: Dripline Structure

Headline result

Halo nuclei (Li-11, Be-11, B-17, etc.) are exotic neutron-rich nuclei near the neutron dripline that feature a “halo” of loosely bound neutrons extending far beyond standard nuclear-shell radius. The halo phenomenon was discovered in the 1980s and remains an active research domain (RIBF at RIKEN, FRIB at Michigan State, RAON in Korea).

BST extends the Mayer-Jensen + $\kappa_{ls} = C_2/n_C = 6/5$ shell-model framework (Vol 3 Ch 2-3) to dripline structure per Task #112 (completed). Halo-nuclei BST predictions:

- Li-11 ($Z = 3, N = 8$): 2-neutron halo; BST identifies halo radius via substrate-mechanism extension
- Be-11 ($Z = 4, N = 7$): 1-neutron halo; ground-state quantum numbers BST-derivable
- B-17 ($Z = 5, N = 12$): proton-rich halo candidate; BST framework consistent

Substrate mechanism

The halo phenomenon emerges from substrate-natural shell-closure weakness at extreme N/Z ratios. Per Vol 3 Ch 2 spin-orbit framework ($\kappa_{ls} = 6/5$) and Vol 3 Ch 3 shell-filling, dripline nuclei have:

1. Last neutron extremely weakly bound: shell-closure margin $\sim$ MeV scale (vs. ~ 10 MeV bulk binding)
2. Substrate-coordinate count for halo region: extended BST-spectral structure beyond N_{max} -scale binding
3. Substrate-natural radial extent: substrate-coordinate Bergman propagation produces “halo” beyond geometric nuclear radius

Per Task #112: BST framework consistent with observed halo-nuclei phenomenology at I-tier (qualitative + structural; quantitative precision multi-week).

Match precision

I-tier framework. Halo-nuclei observables: - Halo-radius BST predictions (existing per Task #112): consistent with measured RMS radii at ~10-30% precision - Ground-state quantum numbers: structurally BST-derivable via Vol 3 Ch 3 extension - Dripline location: BST consistent with experimental dripline mapping

Per Cal #21 STANDING RULE: EMPIRICAL PARTIAL (structural consistency); MECHANISM PATH ARTICULATED via shell-model + substrate extension. D-tier ratification: multi-week per Task #112 follow-on + RIBF/FRIB experimental data.

Cross-volume dependencies

- Vol 3 Ch 2 (Magic Numbers) — $\kappa_{ls} = C_2/n_C$ base
- Vol 3 Ch 3 (Nuclear Shell Model) — 30-isotope BST extension
- Vol 3 Ch 4 (SEMF Coefficients) — bulk binding-energy framework
- Vol 3 Ch 6 (Superheavy Island) — extreme N/Z extension parallel
- Vol 1 Ch 5 (Casimir Algebra) — K-type Casimir extension beyond nuclear bulk

K-audit anchor

K201 Vol 3 Ch 5 Halo Nuclei K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Most nuclei are roughly spherical, with neutrons and protons packed close together. But some nuclei have a “halo” — one or two neutrons that orbit far from the rest of the nucleus, like a moon orbiting at a great distance from a planet. Lithium-11 is the most famous example. BST predicts WHY these halos exist using the same shell-model framework that gives magic numbers. The neutrons in halos are barely bound — they live at the edge of nuclear stability.

Level 2 — Undergraduate physics student

Halo nuclei live near the neutron dripline (the boundary where adding another neutron makes the nucleus unstable). The “halo” describes one or two extra neutrons that orbit at large radius beyond the compact nuclear core. Famous examples: Li-11 (2-neutron halo), Be-11 (1-neutron halo), B-17.

BST framework: extend Mayer-Jensen + $\kappa_{ls} = 6/5$ to dripline regions. Halo emerges when last-bound-neutron binding margin approaches zero. Substrate-natural shell-closure weakness at extreme N/Z makes specific N + Z combinations stable / unstable / halo-prone.

Per Task #112 (completed): BST predictions for halo-radius + ground-state quantum numbers consistent with experimental RIBF/FRIB measurements at I-tier framework precision (~10-30%).

Level 3 — Graduate student / theorem-level

Halo-nuclei structure per BST framework:

1. Bulk binding-energy framework (Vol 3 Ch 4 SEMF): nuclear bulk + surface + Coulomb + asymmetry + pairing terms in BST primary forms $\propto m_p/\pi$
2. Shell-model framework (Vol 3 Ch 2-3): $\kappa_{ls} = C_2/n_C = 6/5$; shell-filling sequence extended to dripline
3. Dripline criterion: $B(A,Z,N+1) - B(A,Z,N) \rightarrow 0$ at neutron dripline; last neutron barely bound
4. Halo emergence: when last-bound neutron has angular momentum $l > 0$ and weakly-bound state, wavefunction extends far beyond nuclear core (asymptotic $1/r$ decay for s-wave; faster decay for higher l but still extended for weakly-bound states)

BST primary forms for halo-relevant observables: - Halo radius RMS $\sim \hbar / \sqrt{2 m_n B_{\text{halo}}}$ where B_{halo} is halo-neutron binding ($\sim$ MeV; substrate-mechanism precision multi-week) - Ground-state J^π for halo nuclei: follows from Vol 3 Ch 3 shell-filling at dripline shells

Per Task #112 (completed): BST framework consistent with observed halo-nuclei at I-tier; substantive precision multi-week per RIBF/FRIB/RAON experimental program.

Cal #21: EMPIRICAL PARTIAL + MECHANISM PATH ARTICULATED $\rightarrow$ I-tier candidate.

Bibliography

1. I. Tanihata et al. (1985): discovery of halo nuclei (Li-11 RMS radius).
2. P. G. Hansen, A. S. Jensen, B. Jonson (1995): review of halo phenomenon.
3. Task #112 (BST repository, completed): halo nuclei structure framework.
4. RIBF (RIKEN), FRIB (MSU), RAON (Korea) experimental programs.
5. Vol 3 Ch 2-4: BST shell-model + SEMF framework.